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ORIGINAL ARTICLE

Screening for diabetic retinopathy with artificial intelligence: a real world evaluation

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Abstract

Aim Periodic screening for diabetic retinopathy (DR) is effective for preventing blindness. Artificial intelligence (AI) systems could be useful for increasing the screening of DR in diabetic patients. The aim of this study was to compare the performance of the DAIRET system in detecting DR to that of ophthalmologists in a real-world setting.

Methods Fundus photography was performed with a nonmydriatic camera in 958 consecutive patients older than 18 years who were affected by diabetes and who were enrolled in the DR screening in the Diabetes and Endocrinology Unit and in the Eye Unit of ULSS8 Berica (Italy) between June 2022 and June 2023. All retinal images were evaluated by DAIRET, which is a machine learning algorithm based on AI. In addition, all the images obtained were analysed by an ophthalmologist who graded the images. The results obtained by DAIRET were compared with those obtained by the ophthalmologist.

Results We included 958 patients, but only 867 (90.5%) patients had retinal images sufficient for evaluation by a human grader. The sensitivity for detecting cases of moderate DR and above was 1 (100%), and the sensitivity for detecting cases of mild DR was 0.84 ± 0.03. The specificity of detecting the absence of DR was lower (0.59 ± 0.04) because of the high number of false-positives.

Conclusion DAIRET showed an optimal sensitivity in detecting all cases of referable DR (moderate DR or above) compared with that of a human grader. On the other hand, the specificity of DAIRET was low because of the high number of false-positives, which limits its cost-effectiveness.

Keywords Diabetic retinopathy screening · Artificial intelligence · Sensitivity · Specificity

Introduction

Diabetic retinopathy (DR) remains one of the leading causes of blindness worldwide in developed countries [1]. Periodic screening for DR has been proven to be effective in preventing blindness, so national and international guidelines recommend annual or biennial screening in diabetic patients [2, 3]. Patients with proliferative diabetic retinopathy (PDR) or macular oedema may be asymptomatic, so only the detection of DR by screening can lead to the initiation of interventions that may prevent vision loss (laser photocoagulation therapy, intravitreal injection of antivascular endothelial growth factor (VEGF)). Fundus photography is considered the gold standard for DR screening [2].

Currently, the proportion of diabetic patients undergoing screening for DR is low. In Italy, in 2022, only 27.1% of type 2 diabetic patients were screened for DR [4]. One of the barriers to increasing the screening of DR lies in the referral to ophthalmologists (which number is often limited) for the purpose of screening, which can cause long wait times and long appointments [5]. Referral to ophthalmologists should be limited to more complex examinations and therapies.

In this context, telemedicine and artificial intelligence (AI) systems could be useful for increasing the screening of DR in diabetic patients.

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Fundus photography with remote readings by ophthalmologists has great potential to increase screening for DR [6]. This type of telemedicine connects diabetologists and ophthalmologists, and it has been demonstrated that this strategy is cost-effective [7].

More recently, the use of AI has also been proposed as a possible solution to enhance DR screening. There are two types of AI systems: machine learning (ML) and deep learning (DL) systems. ML systems are based on automated recognition of DR lesions with high sensitivity in the detection of DR (87–95%) but lower specificity (49.6–68.8%) [8]. These systems are related to a high number of false-positives, which limits their cost-effectiveness.

On the other hand, DL systems require much less human guidance and can be used to detect DR, but they are also used for classifying DR, reducing the percentage of false-positives and enhancing the specificity to 59.4–98% [9, 10]. Recently, the US Food and Drug Administration approved the IDx-DR Service as the first system based on AI that can automatically grade DR with a sensitivity of 87.2% and a specificity of 90.7% for the classification of more than mild DR [11].

In our Diabetes and Endocrinology Unit, in the last two years, we have performed DR screening with the DAIRET algorithm, the commercial Italian version of Retmarker Screening by Retmarker SA, Portugal, in collaboration with the Eye Unit. The DAIRET system has demonstrated high sensitivity in detecting DR (85%) with nonmydriatic images in a Portuguese population [12]. Piatti et al. recently evaluated the accuracy of this algorithm in an Italian population, confirming the excellent sensitivity for referable DR [13].

The aim of this study was to determine the performance of the DAIRET algorithm in detecting DR compared to that of ophthalmologists in a real-world setting in a larger Italian population.

Materials and methods

This was a retrospective and observational study. We analysed data obtained from consecutive patients older than 18 years who were affected by diabetes (type 1 or type 2 or MODY or other type) and who were enrolled in the DR screening in the Diabetes and Endocrinology Unit and in the Eye Unit of ULSS8 Berica (Italy) between June 2022 and June 2023.

The exclusion criteria were age > 75 years, history of laser or intravitreal anti-VEGF treatment, and the presence of any macular disorder.

We also recorded some patients' clinical (age, female/male proportion, type of diabetes, diabetes duration) and metabolic characteristics (HbA1c measured with a standardized, IFCC-aligned HPLC method [14]).

The study was conducted in accordance with the 1964 Declaration of Helsinki and its later amendments and the Good Clinical Practice guidelines. Written informed consent was obtained from all participants.

Fundus photography was performed with a nonmydriatic camera (Nikon Retina Station, Nikon Instruments SPA) by trained nurses or optometrists. All retinal images were 45° in two fields: one centred on the optic disc and the other centred on the macula. Then, all the images were automatically uploaded to Metaclinic (an electronic medical record). DAIRET is a machine learning algorithm based on an artificial intelligence-certified medical device (class II in Europe) that is also approved in Australia, and it is can be used to recognize DR lesions. The DAIRET does not provide a classification of DR, but it is only capable of filtering out the subjects who truly need to be referred to an ophthalmologist. The result of the algorithm is positive screening (which indicates a need for the patient to be evaluated by an ophthalmologist) or negative screening.

All the images obtained were analysed by a single expert, a certified ophthalmologist, who graded the images in a blinded fashion for comparison to the DAIRET results. First, the expert indicated whether the retinal image was of sufficient quality for grading, and then the image was graded following the classification of the International Clinical Diabetic Retinopathy Severity Scale [15]. The DAIRET results were compared with the results of ophthalmologist grading, which is considered the gold standard.

Statistical analysis

The data are expressed as the means ± SDs for continuous variables and percentages for categorical variables. The numbers of patients with mild nonproliferative DR, moderate nonproliferative DR, and severe nonproliferative DR were calculated using the results obtained from human grading (i.e., the expert ophthalmologist).

The DAIRET results were compared with the ophthalmologist results using sensitivity and specificity measures. Sensitivity is expressed as the probability that the DAIRET result was positive when retinopathy was present according to human grading, and specificity is the probability that the DAIRET result was negative when retinopathy was not present according to human grading.

Results

We included 958 patients, but only 867 (90.5%) patients had retinal images sufficient for evaluation by a human grader.

Table 1 Clinical and metabolic characteristics of the study population ($n = 867$)

Variable	
Age (years)	62 ± 13
Men (%)	64.8%
Type 2 diabetic patients (%)	89.2
Type 1 diabetic patients (%)	9
Diabetes duration (years)	13.1 ±. 9.2
HbA1c (mmol/mol)	59 ± 13
HbA1c (%)	7.5 ± 3.3

Table 2 Comparison of the DAIRET results with the results of ophthalmologist grading

		HUMAN GRADER	
		Presence of DR (mild, moderate or above)	Absence of DR
DAIRET RESULTS	Positive	292	214
	Negative	53	308
	TOTAL	345	522

For the clinical characteristics (Table 1), 64.8% of the patients were men, their mean age was 62 ± 13 years, 89.2% were affected by type 2 diabetes, and 9% were affected by type 1 diabetes; finally, the mean diabetes duration was 13.1 ±. 9.2 years. The mean HbA1c value was 59 ± 13 mmol/mol (7.5 ± 3.3%).

As shown in Tables 2 and 522 patients had no DR, 331 patients had mild nonproliferative DR (38.3% of the sample), and 14 patients had moderate nonproliferative DR or worse (1.6% of the sample).

In the 522 patients without DR, in 214 patients, the DAIRET result did not match the human grader result; in particular, the DAIRET result was a positive screening, but the human grader result was no DR (i.e., false-positive DAIRET result). Most of these phenomena were due to image artefacts (57% of false-positives); in the other cases, they were due to other ocular diseases, such as glaucoma, age-related macular degeneration, and choroidal nevus (43% of false-positives).

Table 3 shows the calculated accuracy measures. In particular, a comparison of the DAIRET and human grader results revealed few false-negative test results (53/331 cases; 16% false-negatives) but a greater percentage of false-positive results (214/522 cases; 40.9% false-positives).

The sensitivity for detecting 14 patients with moderate DR and above was 1 (100%), and the sensitivity for detecting 331 patients with mild DR was 0.84 (95% CI ± 0.03). The specificity for detecting the absence of DR (522 patients) was 0.59 (95% CI ± 0.04), indicating a high percentage of false-positives (40.9% false-positives).

Discussion

In this study, we demonstrated in a real-world setting that the DAIRET system achieved optimal sensitivity in detecting all cases of referable DR (moderate DR or above) compared with that of a human grader in a large diabetic population.

In 1990, representatives of several European countries stated the goal of screening at least 80% of the diabetic population for DR [16]. Unfortunately, at present, the proportion of diabetic patients undergoing screening for DR is very low. The number of ophthalmologists is limited because of the long wait times and long appointment times. The use of AI systems can be a possible solution to increase the proportion of the diabetic population screened for DR.

Considering this context, in our study, the DAIRET system detected all patients who needed ophthalmologist evaluation or treatment (moderate DR or more). This is an excellent result that was recently confirmed in data published in a smaller population diabetic patients [13]. The sensitivity in detecting cases of mild DR decreased to 84%, but these data support the safety of the system because in these forms of DR, the ophthalmologist is not involved in the treatment, and the only treatment is to improve metabolic control, which must be the principal objective for the diabetologist in all diabetic patients. Notably, in our study, the sensitivity for detecting mild DR was greater than that obtained by Piatti et al. [13] (84% vs. 69%), so this is an encouraging result for the large-scale use of the DAIRET system. These data underscore the possibility of using the DAIRET system in clinical practice to improve DR screening in diabetic patients, as all patients who need treatment and care by ophthalmologists can be detected. This is the most important finding from this study, as it supports previous findings [13] and provides clinicians with confidence in the safety of the DAIRET system. In addition, these data underline the effectiveness of the DAIRET system because it defines patients as either referable or not referable to an ophthalmologist.

On the other hand, in our study, the DAIRET system demonstrated low specificity because of the high number of false-positives (40.9%). These data contrast with recently published data [13] but are in line with the percentage of false-positives observed with the use of other ML systems

Table 3 Sensitivity and specificity (95% CI) analysis comparing the DAIRET results and the human grader results

DR grade	Human grader	DAIRET	DAIRET specificity	FP (%)
No DR	522	308	0.59 (0.55 ± 0.64)	214 (40.9%)
Mild DR	331	278	0.84 (0.81 ± 0.87)	53 (16.0%)
Moderate DR or more	14	14	1	0

(49.6–68.8%) [8]. These data limit the cost-effectiveness of screening with the DAIRET system because all these patients are unnecessarily evaluated by an ophthalmologist. Piatti et al. reported that the false-positive rate was principally due to the foveal reflex in type 1 diabetic patients [13], suggesting that this system is not suitable for these patients. In our study, most of the false-positives were due to imaging artefacts and ocular disease (glaucoma, age-related macular degeneration, choroidal nevus). An improvement of the algorithm is needed to achieve two goals: recognition by the system of image artefacts (perhaps in addition to the use of a specific nonmydriatic camera) and recognition of ocular lesions not due to DR to reduce the false-positive rate. Only with the reduction in the false-positive rate will it be possible to enhance the cost-effectiveness of DR screening with the DAIRET system and its use in large-scale screening.

Considering all our results, the DAIRET system was demonstrated to be an efficient solution for improving the sensitivity of the DR screening system to referable DR and for reducing the need for ophthalmologist intervention.

However, at present, AI systems can help but cannot replace human intervention, and they have medicolegal implications. Recently, in Italy, a list of recommendations regarding the use of AI in health care systems has been published [17]; in particular, it has been established that human intervention is needed to validate AI system results. Currently, this is a barrier to increasing the use of AI systems in daily practice. In particular, all the DAIRET results need to be validated by human intervention, so at the moment, all the screening results are evaluated by specialists (positive and negative screening), resulting in reduced cost-effectiveness of the AI system. In addition, it is important to emphasize that the key point is to define who validates the results (Diabetologists? Ophthalmologists?) in clinical practice. To enhance the cost-effectiveness of DR screening with AI systems such as the DAIRET, it would be necessary to have national and international guidelines that better define the limits of the use of AI in health care systems.

The strength of this study is the large population evaluated in a real-world setting. The main limitation is the use of only one ophthalmologist as a human grader.

In conclusion, in our study, the DAIRET system achieved optimal sensitivity in detecting DR and needs to be evaluated by an ophthalmologist in a large diabetic population. Currently, two main issues remain: the high number of false-positives, mainly due to image artefacts and other ophthalmic disorders, and the medicolegal aspect (the AI system can help but cannot replace human intervention). The technology is evolving rapidly, so it is desirable that the technical issues be improved soon. It is also desirable that national and international guidelines include the use of AI systems to enhance DR screening since there are many studies in the literature regarding the safety and effectiveness of these methods [8–10]. Certainly, multicentre, randomized and real-world studies need to be conducted to use AI systems for DR screening in diabetic patients in daily clinical practice.

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Author contributions All the authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by Silvia Burlina, Sandra Radin, Marzia Poggiato, and Simonetta Lombardi. The first draft of the manuscript was written by Silvia Burlina, and all the authors commented on previous versions of the manuscript. All the authors have read and approved the final manuscript.

Declarations

Conflict of interest The authors declare that they have no conflicts of interest.

Ethics approval Approval was obtained from the ethics committee of ULSS8 Berica. The procedures used in this study adhered to the tenets of the Declaration of Helsinki.

Informed consent Informed consent was obtained from all individual participants included in the study.

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